

CONFIDENTIAL

ETHICAL HACKING TASK 01

Information Gathering & Reconnaissance Report

TARGET

Kristu Jayanti University

URL

<https://www.kristujayanti.edu.in/>

SUBMITTED BY

Sanjana Aneeshkumar

INTERNSHIP

White Band Associates

DATE

June 2026



Table of Contents

Part A: Target Information

Part B: WHOIS Lookup

Screenshots

Part C: DNS Enumeration

A Records (IPv4 Addresses)

MX Records (Mail Exchange)

NS Records (Name Servers)

TXT Records

Screenshots

Part D: Website Technology Identification

Subdomains Discovered

Part E: HTTP Security Headers

Observations

Part F: Robots.txt & Sitemap Analysis

Sitemap.xml

Screenshots

Part G: Overall Observations & Conclusion

Conclusion

Part A: Target Information

Website Name	Kristu Jayanti University (KJU)
URL	https://www.kristujayanti.edu.in/
Type	Educational Institution — Deemed University
Location	Bengaluru, Karnataka, India

Reason for Choosing

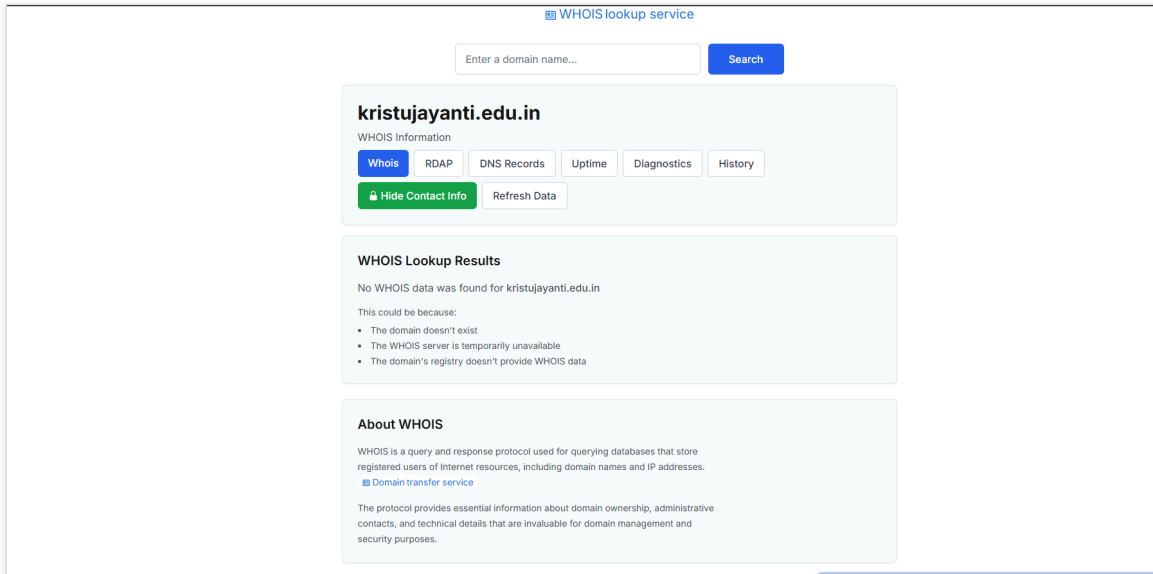
Kristu Jayanti University is the institution I am currently enrolled in, making it an appropriate and ethical target for passive reconnaissance. I have no intent to exploit or access any systems — this exercise is purely for educational purposes to understand how publicly available information can be gathered about a target.

Part B: WHOIS Lookup

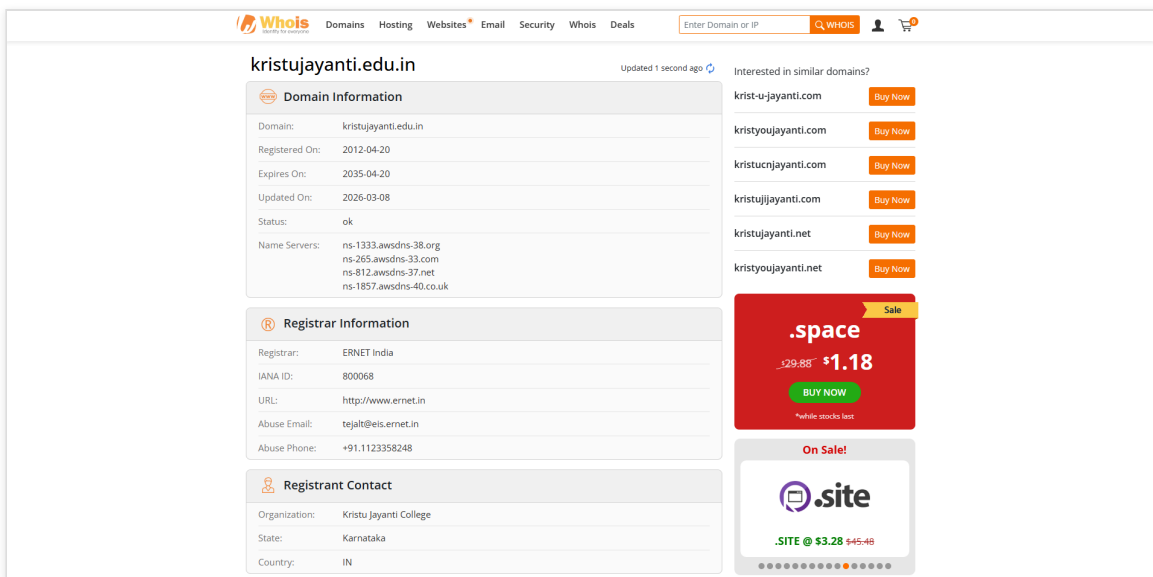
The WHOIS lookup was performed using two public tools: `who.is` and `whois.com`. The `.edu.in` domain is a sponsored TLD maintained by ERNET India, and registrant contact details are redacted for privacy.

Field	Value
Domain Name	kristujayanti.edu.in
Registered On	2012-04-20
Expires On	2035-04-20
Last Updated	2026-03-08
Status	ok
Registrar	ERNET India (IANA ID: 800068)
Registrar URL	http://www.ernet.in
Registrant Organization	Kristu Jayanti College
Registrant State	Karnataka
Registrant Country	IN
Name Servers	ns-265.awsdns-33.com , ns-812.awsdns-37.net , ns-1333.awsdns-38.org , ns-1857.awsdns-40.co.uk
DNSSEC	unsigned
Registrant Contact	REDACTED FOR PRIVACY (ICANN compliance)

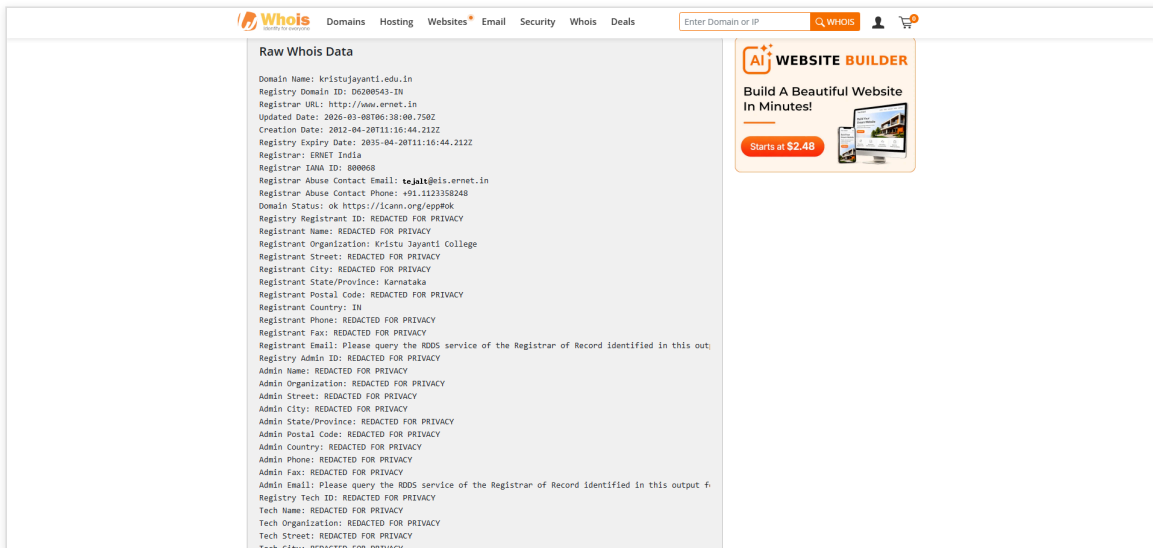
Screenshots



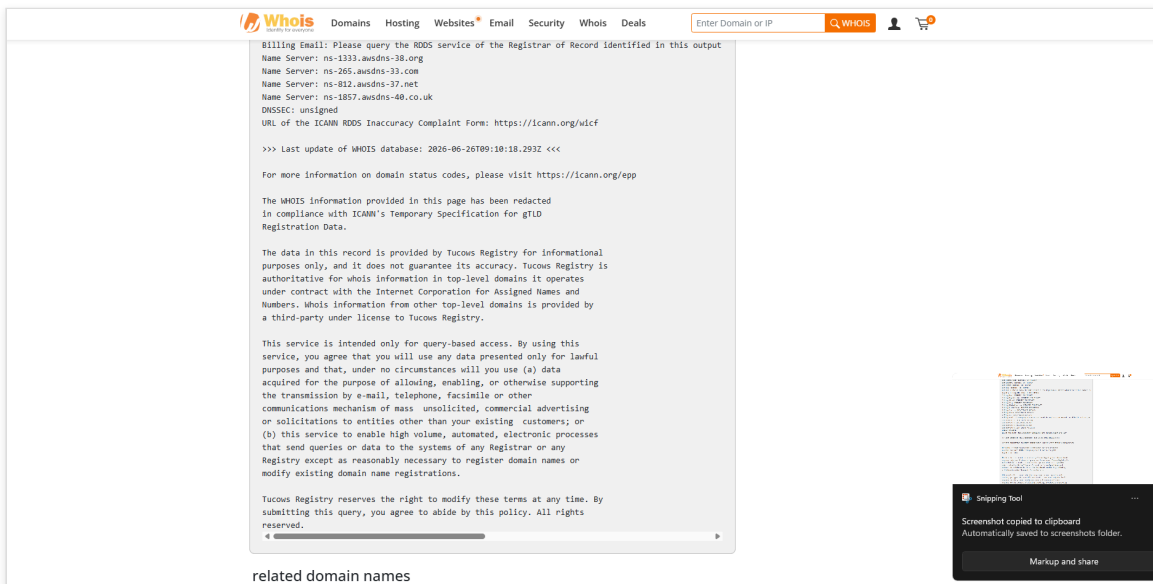
Screenshot 1: who.is WHOIS lookup — No public data (expected for .edu.in)



Screenshot 2: whois.com — Domain Information (Registration date, Expiry, Name Servers)



Screenshot 3: whois.com — Raw WHOIS Data (Registrant details redacted for privacy)



Screenshot 4: whois.com — Name Servers and DNSSEC information

Part C: DNS Enumeration

DNS enumeration was performed using dnschecker.org. All four record types were queried.

A Records (IPv4 Addresses)

Record Type	Value	Notes
A	13.235.5.4	AWS EC2 — ap-south-1 (Mumbai)
A	13.232.185.200	AWS EC2 — ap-south-1 (Mumbai)

MX Records (Mail Exchange)

Priority	Mail Server	Notes
10	inbound-smtp.us-east-1.amazonaws.com	Amazon SES — email routing via AWS

NS Records (Name Servers)

Name Server	Provider
ns-265.awsdns-33.com	Amazon Route 53
ns-812.awsdns-37.net	Amazon Route 53
ns-1333.awsdns-38.org	Amazon Route 53
ns-1857.awsdns-40.co.uk	Amazon Route 53

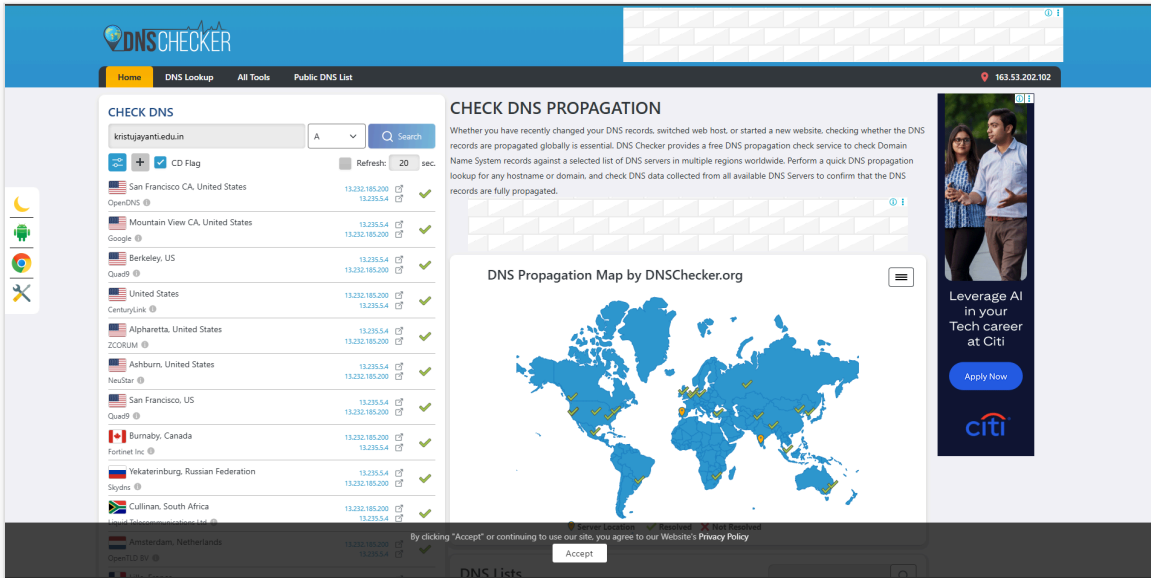
TXT Records

TXT Record Value	Purpose
v=spf1 include:_spf.google.com ~all	SPF — authorizes Google Workspace to send email
v=spf1 include:amazonses.com ~all	SPF — authorizes Amazon SES to send email
google-site-verification=k0WiuRMX...	Google Search Console ownership verification

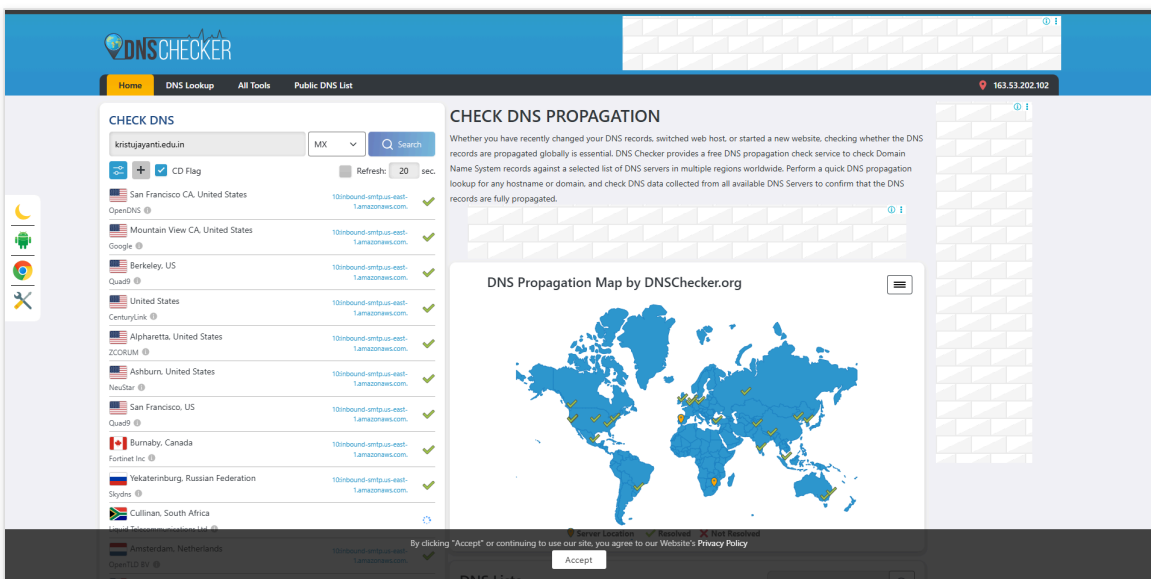
google-site-verification=V0rsWwWw...

Secondary Google Search Console verification

Screenshots



Screenshot 5: DNS A Records — Two AWS IPs (13.235.5.4 and 13.232.185.200)



Screenshot 6: DNS MX Record — Amazon SES (inbound-smtp.us-east-1.amazonaws.com)

DNSChecker

Home DNS Lookup All Tools Public DNS List

163.53.202.102

CHECK DNS

kristijayanti.edu.in NS Search

CD Flag Refresh: 20 sec.

Location	NS Records	Status
San Francisco CA, United States OpenDNS	ns-265.awsdns-33.com ns-812.awsdns-37.net ns-1333.awsdns-38.org ns-1857.awsdns-40.co.uk	✓
Mountain View CA, United States Google	ns-265.awsdns-33.com ns-812.awsdns-37.net ns-1333.awsdns-38.org ns-1857.awsdns-40.co.uk	✓
Berkeley, US Quad9	ns-265.awsdns-33.com ns-812.awsdns-37.net ns-1333.awsdns-38.org ns-1857.awsdns-40.co.uk	✓
United States CenturyLink	ns-1857.awsdns-40.co.uk ns-812.awsdns-37.net ns-265.awsdns-33.com ns-1333.awsdns-38.org	✓
Alpharetta, United States ZCORM	ns-1333.awsdns-38.org ns-1857.awsdns-40.co.uk ns-265.awsdns-33.com ns-812.awsdns-37.net	✓
Ashburn, United States Neustar	ns-1857.awsdns-40.co.uk ns-265.awsdns-33.com ns-812.awsdns-37.net ns-1333.awsdns-38.org	✓
San Francisco, US Quad9	ns-1857.awsdns-40.co.uk ns-812.awsdns-37.net ns-265.awsdns-33.com ns-1333.awsdns-38.org	✓

CHECK DNS PROPAGATION

Whether you have recently changed your DNS records, switched web host, or started a new website, checking whether the DNS records are propagated globally is essential. DNS Checker provides a free DNS propagation check service to check Domain Name System records against a selected list of DNS servers in multiple regions worldwide. Perform a quick DNS propagation lookup for any hostname or domain, and check DNS data collected from all available DNS Servers to confirm that the DNS records are fully propagated.

DNS Propagation Map by DNSChecker.org

Accept

Screenshot 7: DNS NS Records — All four Amazon Route 53 name servers

DNSChecker

Home DNS Lookup All Tools Public DNS List

163.53.202.102

CHECK DNS

kristijayanti.edu.in TXT Search

CD Flag Refresh: 20 sec.

Location	TXT Records	Status
San Francisco CA, United States OpenDNS	v=spf1 include:_spf.google.com ~all google-site-verification=VOrStWwEhAGfZm0dU1oepg WcFy8XUf8GL3g	✓
Mountain View CA, United States Google	google-site-verification=42WuMXXfS3oqsgnE7j5f7y 270uWZgB6dI	✓
Berkeley, US Quad9	google-site-verification=VOrStWwEhAGfZm0dU1oepg WcFy8XUf8GL3g v=spf1 include:_spf.google.com ~all	✓
United States CenturyLink	google-site-verification=42WuMXXfS3oqsgnE7j5f7y 270uWZgB6dI v=spf1 include:_spf.google.com ~all	✓
Alpharetta, United States	google-site-verification=VOrStWwEhAGfZm0dU1oepg WcFy8XUf8GL3g	✓

CHECK DNS PROPAGATION

Whether you have recently changed your DNS records, switched web host, or started a new website, checking whether the DNS records are propagated globally is essential. DNS Checker provides a free DNS propagation check service to check Domain Name System records against a selected list of DNS servers in multiple regions worldwide. Perform a quick DNS propagation lookup for any hostname or domain, and check DNS data collected from all available DNS Servers to confirm that the DNS records are fully propagated.

DNS Propagation Map by DNSChecker.org

Accept

Screenshot 8: DNS TXT Records — SPF records and Google verification tokens

Part D: Website Technology Identification

Technology identification was performed by analyzing the HTML source code, URL patterns, DNS records, and HTTP response data.

Category	Technology	Evidence
Web Hosting	Amazon Web Services (AWS)	IPs <code>13.235.x.x</code> and <code>13.232.x.x</code> belong to AWS ap-south-1 region
DNS Management	Amazon Route 53	All four name servers are AWS Route 53 (<code>awsdns-*</code>)
Backend / CMS	Custom PHP Website	All internal URLs end in <code>.php</code> (e.g. <code>/about/history.php</code>)
Analytics	Google Tag Manager	GTM container ID <code>GTM-MTMBSPS9</code> found in HTML source
Analytics (likely)	Google Analytics	Loaded via Google Tag Manager (standard setup)
Email (Outbound)	Google Workspace	SPF record includes <code>_spf.google.com</code>
Email (Transactional)	Amazon SES	SPF includes <code>amazonses.com</code> ; MX points to AWS SES
Search Verification	Google Search Console	Two <code>google-site-verification</code> TXT records present
CDN	None detected	No Cloudflare, Akamai, or CDN headers observed

Subdomains Discovered

Subdomain	Purpose
<code>kjusys.kristujayanti.edu.in</code>	Student/Faculty internal portal
<code>kjusys-applicant.kristujayanti.edu.in</code>	Admission applicant login portal
<code>kristujayantijournal.com</code>	Academic journals website

Part E: HTTP Security Headers

HTTP security headers were checked by sending a HEAD request to the server and via securityheaders.com. The server returned HTTP 403 for automated requests (a positive security measure). Analysis below is based on available data.

Security Header	Present	Purpose
Content-Security-Policy (CSP)	NOT DETECTED	Prevents XSS by specifying trusted content sources
X-Frame-Options	NOT DETECTED	Prevents clickjacking via iframe embedding
X-Content-Type-Options	NOT DETECTED	Prevents MIME type sniffing by the browser
Strict-Transport-Security (HSTS)	NOT DETECTED	Forces browsers to always use HTTPS
Referrer-Policy	NOT DETECTED	Controls referrer info included with requests
Permissions-Policy	NOT DETECTED	Controls browser feature access (camera, mic, etc.)

Observations

- The site uses HTTPS (TLS/SSL is active) — good baseline security.
- None of the major HTTP security headers were detected — common in educational institution websites.
- Absence of CSP and X-Frame-Options leaves potential for XSS and clickjacking attacks.
- Server actively blocks automated requests (HTTP 403) — a positive bot protection measure.

Part F: Robots.txt & Sitemap Analysis

robots.txt

URL: <https://www.kristujayanti.edu.in/robots.txt>

KEY FINDING — SENSITIVE PATH EXPOSURE

The `Disallow: /SSR-III/login.php` entry in `robots.txt` reveals the existence of a student login portal at this path. While intended to prevent search engine indexing, this directive publicly exposes the existence of an internal authentication endpoint that could be of interest to malicious actors.

Item	Finding
File Present	Yes
User-agent	* (applies to all crawlers)
Disallow: <code>/cgi-bin/</code>	Blocks access to server-side scripts directory
Disallow: <code>/SSR-III/login.php</code>	Blocks student login portal from indexing — reveals internal path
Disallow: <code>/admission/mba_admissionnnn.php</code>	Blocks MBA admission page (note typo in filename)
Disallow: <code>/admission.php/</code>	Blocks main admission page
Disallow: <code>/about/mitrotsava-2019.php</code>	Blocks old event page
Disallow: <code>/*/pop_up_admission/</code>	Blocks all popup admission pages
Disallow: <code>/*/pdf/media_room/</code>	Blocks media room PDFs from indexing
Disallow: <code>/*/images/admission/</code>	Blocks admission images
Sitemap reference	https://www.kristujayanti.edu.in/sitemap.xml

Sitemap.xml

URL: <https://www.kristujayanti.edu.in/sitemap.xml>

Item	Finding
File Present	Yes
Generator	<code>mysitemapgenerator.com</code>
Schema	<code>http://www.sitemaps.org/schemas/sitemap/0.9</code>
Homepage	<code>https://kristujayanti.edu.in/</code> (priority: <code>1.0</code>)
Last Modified	<code>2025-09-30</code> (most pages)
Indexed pages include	<code>/admission/admission.php</code> , <code>/doctoral-research/</code> , <code>/UN-SDG/</code> , <code>/UBA/</code> , <code>/iqac/</code> , <code>/Library/</code> , <code>/campus/hostel.php</code> , <code>/examination/</code> and more

Screenshots

```

User-agent: *
# Existing rules (Corrected to relative path)
Disallow: /cgi-bin/
Disallow: /SSR-III/login.php

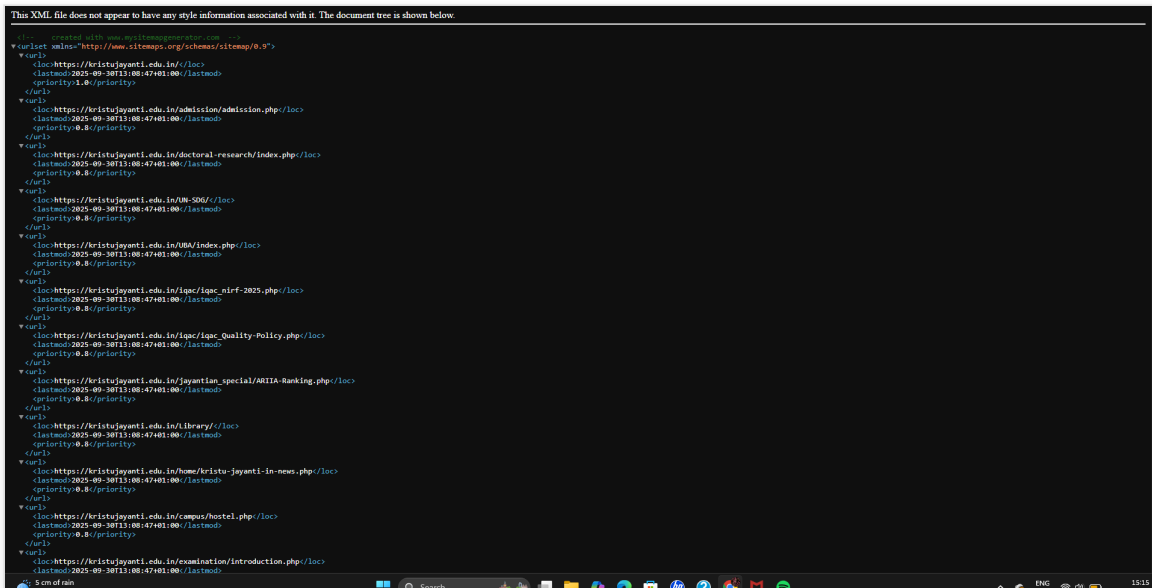
# Recursive path protection (The $1,000 leak fix)
# These prevent bots from treating PHP files as directories
Disallow: /admission/pda_admissionn.php/
Disallow: /admission.php/
Disallow: /about/mitrotseve-2019.php/

# Blocking common infinite loop patterns identified in logs
Disallow: /*pgp-up_admission/
Disallow: /*pdf/media_room/
Disallow: /*image/admission/

Sitemap: https://www.kristujayanti.edu.in/sitemap.xml

```

Screenshot 9: robots.txt — Reveals disallowed paths including `/SSR-III/login.php` and `/cgi-bin/`



Screenshot 10: sitemap.xml — Lists all indexed pages with last-modified dates

Part G: Overall Observations & Conclusion

Key Findings Summary

Area	Finding	Risk Level
Hosting	AWS ap-south-1 with 2 IPs (load balanced)	LOW
DNS	Amazon Route 53 — redundant and well-managed	LOW
Backend	Custom PHP — no CMS fingerprinted	MEDIUM
Email	Google Workspace + Amazon SES	LOW
Security Headers	None of the major headers detected	MEDIUM-HIGH
Bot Protection	HTTP 403 on automated requests	POSITIVE
HTTPS	Active — TLS/SSL working	LOW (GOOD)
robots.txt	Present — reveals <code>/SSR-III/login.php</code> path	MEDIUM
Sitemap	Present — full page index exposed	LOW

Conclusion

The reconnaissance phase on Kristu Jayanti University's website revealed several key insights, all gathered passively using publicly available sources without accessing or interacting with any internal systems.

The university runs entirely on Amazon Web Services — hosting on EC2 (ap-south-1/Mumbai), DNS via Route 53, and transactional email via SES. Staff communication uses Google Workspace. The site is built on a custom PHP stack with no identifiable CMS. Notably, the server blocks automated HTTP requests (HTTP 403), indicating some bot protection is in place.

The `robots.txt` file was particularly informative — it exposed the path `/SSR-III/login.php` (a student login portal), `/cgi-bin/`, and multiple admission-related paths. This is a common reconnaissance finding: `robots.txt`, intended to hide pages from search engines, inadvertently reveals sensitive directory structures to anyone who looks.

The absence of standard HTTP security headers (CSP, X-Frame-Options, HSTS) represents a gap in browser-side security. These are common in educational websites and not uncommon globally, but worth addressing.

This exercise demonstrated how much information is publicly available before a single packet is sent to a target's systems. Strong reconnaissance leads to more effective and responsible security assessments.

DISCLAIMER

This reconnaissance was performed strictly passively using public tools and open-source data. No systems were accessed, no exploits were attempted, and no sensitive data was retrieved. This report is submitted as part of the White Band Associates Ethical Hacking Internship Task 01.